PeopleLens — Workforce Intelligence

Turning workforce data into better business decisions

An end-to-end people analytics product: data engineering → SQL analytics → statistics → forecasting & machine learning → decision support.

Live app: **peoplelens.streamlit.app** · Code: **github.com/ayushkhandelwal-bits/peoplelens** · **LinkedIn** · **ayushkhandelwal.connect@gmail.com**

THE BUSINESS PROBLEM

People data sits in separate systems — HRMS, recruitment, surveys, payroll, budgets — so leaders get headcount *reports* but not answers: **Will we have the capacity we planned? Where is attrition genuinely abnormal? What does it cost? Which levers are worth pulling?** PeopleLens integrates these sources into one validated model and answers with stated uncertainty.

SCALE (SYNTHETIC, FICTIONAL ORGANISATION)

9,336 employees over 36 months	**545k** source rows in 16 tables	**138** data-quality rules, all pass
35 KPIs, one definition each	**6** forecast models compared	**102** automated tests

HOW IT WORKS

1 · Simulate — Month-by-month workforce: exits, hiring funnel, promotions, pay, surveys (seed 42)

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2 · Engineer — HRMS-style export with checksums → staging → cleaning → quality gate

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3 · Store — PostgreSQL / SQLite, effective-dated history, monthly mart, role-restricted views

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4 · Analyse — Metric layer, significance tests, survival curves, forecasts, attrition model, NLP

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5 · Decide — Streamlit app, what-if simulators, recommendations, monthly report, plain-English Q&A

KEY FINDINGS

- **One real hotspot, not 24:** Customer Support attrition 30.2% vs 18.6% overall (1.75× leaver rate); only 2 of 24 departments differ after multiple-comparison correction.
- **Location is a red herring:** city differences vanish once department mix is controlled (p = 0.73) — target teams, not cities.
- **Volume ≠ value in sourcing:** referrals convert 13.8% of applicants with 96.6% 90-day retention; the biggest channel converts 4.7%.
- **Behind plan:** 6,106 vs 6,195 planned (−89): attrition above the plan's assumption plus 2–4-month time-to-fill.
- **Cost of turnover:** 920 voluntary exits ≈ ₹39.8 Cr a year (sensitivity ₹29.8–54.6 Cr).

MODELS — TESTED ON LATER PERIODS, AGAINST BASELINES

Question	Approach	Result
Headcount, 6 months	Stock-and-flow model vs Naive, Drift, Holt, Holt-Winters, SARIMA	MAE 29 heads vs 97 naive; SARIMA a tie (28.4) → explainable flow model kept
Who may resign	Logistic regression vs gradient boosting	AUC 0.64, top-decile lift 1.9× — team-level use only
Time to fill	Kaplan-Meier vs point models	No model beat the median → plan with P50/P80 ranges
What-ifs	Driver simulator on the attrition model; scenarios	Levers → leavers, headcount, turnover and pay cost

GOVERNANCE & RESPONSIBLE USE

- Quality gate blocks publication on any error; full lineage and run log.
- Analyst / Viewer access enforced by the **database**, not just the screen.
- Groups under 10 suppressed; comments carry no employee ID; no gender in models.
- Refuses firing or disciplinary questions — models guide team conversations, never individual decisions.

STACK — OPEN SOURCE, ZERO COST

Python · pandas · NumPy · SQL · PostgreSQL · SQLite · SQLAlchemy · statsmodels · SciPy · scikit-learn · VADER · TF-IDF · Streamlit · Plotly · pytest · Ruff · Docker · Playwright

Ask PeopleLens answers plain-English questions with tested SQL templates — no paid AI API.

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All data is synthetic for a fictional organisation; findings illustrate the method, not a real company.